The impact of punctuated infectiousness on infectious disease dynamics

Author One[1], Author Two[1,2], Author Three[2]

[1]Department of Computer Science, University of Colorado Boulder, Boulder CO USA
[2]BioFrontiers Institute, University of Colorado Boulder, Boulder CO USA
[3]Department of Epidemiology, Colorado School of Public Health, Aurora CO USA

Abstract

Narrative outline (working document — remove before submission)

Core argument in one paragraph

The population-level infectiousness profile $A(\tau)$ is an expectation over individuals, and many different individual-level profiles $a_i(\tau)$ can produce the same $A(\tau)$. The "punctuatedness" of $a_i(\tau)$ — how concentrated in time each person's infectiousness is — is a free parameter (ψ) that is invisible to standard surveillance but controls stochastic epidemic dynamics. We introduce a framework for studying this, show that ψ affects epidemic timing, overdispersion, and intervention effectiveness, and discuss how ψ might be inferred.

---

1. Introduction

Message: The individual infectiousness profile $a_i(\tau)$ is a fundamental but neglected object. Different $a_i(\tau)$ can yield the same $A(\tau)$, so standard methods cannot distinguish them — but they may matter for epidemic dynamics. This paper asks: do they matter, and how would we know?

Structure (existing, largely in place):

1. $A(\tau) = R_0 g(\tau)$ is the fundamental object driving epidemic dynamics.
2. $A(\tau) = E_i[a_i(\tau)]$: the population profile is an average over individuals.
3. Since A is an average, many different a_i can produce the same A. SEIR example: stepwise, smooth, and spike profiles all yield the same bell-shaped A (Figure 1A–C).
4. We don't know what $a_i(\tau)$ looks like in general, and it's hard to measure.
5. But there are good reasons to think it matters: it interacts with contact patterns, detection timing, and early-outbreak stochasticity.
6. Our goals: (a) introduce a framework for parameterizing a_i's "punctuatedness" ψ, (b) characterize how ψ impacts epidemic dynamics holding $A(\tau)$ fixed, (c) assess identifiability of ψ.

---

2. Model

Message: The time-shifted Gamma family provides a one-parameter continuum between spike and smooth individual infectiousness profiles, all sharing the same population-level $A(\tau)$. This is the tool that lets us isolate the effect of punctuatedness.

Structure (existing, largely in place):

1. The decomposition $a_i(\tau) = \nu_i\, b_i(\tau)\, c_i(t_i + \tau)$: biological intensity ($\nu_i$), biological timing ($b_i$), and contact modulation (c_i). This decomposition separates *how much* from *when* from *who's around*.

2. The time-shifted Gamma model for $b_i(\tau)$: latent period $l_i \sim \text{Gamma}((1-\psi)\alpha, \beta)$, then biological profile $f_\epsilon \sim \text{Gamma}(\psi\alpha, \beta)$. Gamma additivity guarantees $E_i[b_i] = g(\tau)$ for any ψ.
3. Contact models: the **reference** process ($c_i \equiv 1$), plus two models used as test beds when needed:
   - *Periodic*: $c_i(t) = 1 - A\cos(\omega t)$, capturing diurnal contact rhythms. [Introduced briefly; the reader should see this as "the simplest nontrivial contact variation."]
   - *Lomax–Poisson*: piecewise-constant with heavy-tailed levels. [Introduced briefly; used only in simulations to show robustness.]

Key framing note: The contact models are *supporting actors*, not co-equal model components. They exist to create settings where ψ can bite. Keep their exposition short (1–2 paragraphs each) and position them as "we need *some* source of contact variation; here are the simplest choices."

---

3. Uncontrolled epidemics

Message: Even in uncontrolled epidemics, ψ shapes observable features of outbreak dynamics. The effects escalate with model complexity: first timing, then overdispersion.

Structure — three subsections of increasing complexity:

1. **Epidemic timing (constant contacts, $c_i \equiv 1$).**
   Punchline: Narrower $a_i \Rightarrow$ later and more variable epidemic establishment.
   Content: The random time shift ς from branching process theory (Morris et al. 2024). Closed-form expressions for $\text{Var}[W]$ and $\text{Mode}[\varsigma]$ as functions of ψ (Eqs. in current draft). ψ enters only through $(((1+z)^2/(1+2z))^{(1-\psi)\alpha}$. Amplified by R_0. Simulations confirm.
   Deliverables: Survival curves for time-to-100 infections (influenza, omicron, measles $\times$ spike/smooth).
   Status: Analytically complete; simulations and figure needed.
2. **Growth rate estimation uncertainty (constant contacts).**
   Punchline: Narrower $a_i \Rightarrow$ more uncertain empirical estimates of r.
   Content: Brief simulation-based result. Logically follows from the time-shift variance result.
   Deliverables: Box-and-whisker or density plots of estimated r across ψ.
   Status: Simulations needed.
3. **Overdispersion from contact variation.**
   Punchline: When contacts vary over time, narrower a_i "see" more of that variation, generating overdispersion in secondary infections — a new mechanism for superspreading that doesn't require individual-level differences in infectiousness or contacts.
   Content:
   - *Setup (2–3 sentences):* With $c_i \equiv 1$, there's no overdispersion regardless of ψ. But real contacts aren't constant. What happens when we add temporal contact variation?
   - *Analytic result (periodic model):* $k = (2/A^2)(1+\omega^2/\beta^2)^{\psi\alpha}$. State the formula, interpret the two factors (contact amplitude $\times$ Fourier averaging), note monotonicity in ψ. Reference Supplementary Methods for derivation.
   - *Simulation confirmation (Lomax model):* Same qualitative pattern under realistic heavy-tailed contacts. One sentence + figure panel.

- *General principle (optional, one sentence):* For any stationary contact process with spectral density $S_c(\omega)$, overdispersion equals the spectral overlap between contact noise and the profile window.

Deliverables: k vs. ψ for the periodic model (analytic curve + simulation dots); a panel showing the same under Lomax contacts.
Status: Analytic result and derivation complete; simulations needed.

---

4. Controlled epidemics

Message: ψ doesn't just affect "background" stochastic properties — it changes how interventions work, in ways that can't be captured by a single effectiveness number.

Structure:

1. **Detect-and-isolate interventions.**
Punchline: Test efficacy (TE) depends on ψ when detection is anchored to infectiousness timing. But TE is only part of the story: isolation also reshapes $g(\tau)$ and increases overdispersion, both ψ-dependently.
Content:
   - TE for symptom-based detection (deterministic offset, then random offset): narrow profiles make detection an all-or-nothing proposition; smooth profiles degrade gracefully.
   - TE for regular testing: sensitivity to screening frequency depends on ψ.
   - Beyond TE: isolation shortens generation intervals (more for smooth) and increases overdispersion (more for spike). Naive simulations using $R_{\text{eff}} = R_0(1 - \text{TE})$ miss these effects.
   - Epidemic simulations illustrating the discrepancy.

   Deliverables: TE curves vs. detection offset (for spike/intermediate/smooth); $g(\tau)$ and k under isolation; paired epidemic trajectories (naive R_{eff} vs. explicit isolation model).
   Status: Framework described; analytics and simulations needed.
2. **Gathering size restrictions.**
Punchline: Gathering size caps interact with ψ in a potentially counter-intuitive way: they reduce mean transmission for everyone, but smooth profiles lose more overdispersion (which was suppressing outbreak probability for spike profiles).
Content: Brief analytic or simulation-based treatment.
Deliverables: Outbreak probability and/or effective k under gathering caps vs. ψ.
Status: Conceptual; needs development.

---

5. Identifiability and inference

Message: ψ is hard to estimate from epidemic data alone, but identifiable from sufficiently detailed contact-tracing data — especially when R_0 is high (which is exactly when ψ matters most).

Structure:

1. When is ψ identifiable? Conditions on data quality (precise infection times, known contact histories).
2. Bayesian inference framework for ψ given contact-tracing data. Knowing the "negatives" (contacts that didn't transmit) dramatically improves power.
3. Power analysis: sample sizes needed to distinguish ψ values, as a function of R_0.

Deliverables: Posterior distributions for ψ under simulated data; power curves.
Status: Framework described; implementation needed.

---

6. Discussion

Key points to make:

1. The individual infectiousness profile is a "hidden variable" that shapes observable epidemic features. Standard surveillance data (R_0, $g(\tau)$) cannot distinguish it, but it controls stochastic dynamics, intervention effectiveness, and even the interpretation of superspreading.
2. The "temporal superspreading" mechanism is novel: overdispersion arising not from who is more infectious, but from *when* infectiousness falls relative to contact variation. There is no "superspreader" to identify — it's a timing coincidence.
3. Intervention planning requires knowledge of $a_i(\tau)$: TE alone is insufficient because isolation reshapes $g(\tau)$ and k in ψ-dependent ways, leading to dynamics that a naive R_{eff} calculation cannot predict.
4. Identifiability of ψ is better when it matters most (high R_0), which is encouraging.
5. Limitations: we assume a shared ψ (but the Gamma additivity holds for heterogeneous ψ_i); we don't model individual-level variation in ν_i jointly with ψ; we focus on respiratory-like pathogens with unimodal kernels.

Figures (planned)

1. **Schematic** (3×3): smooth/stepwise/spike $a_i(\tau)$ (A–C); cumulative infections (D–F); daily infections (G–I). [Partially complete in Mathematica.]
2. **Uncontrolled — epidemic timing:** Survival curves for time-to-100 infections across ψ and R_0.
3. **Uncontrolled — overdispersion:** k vs. ψ (periodic analytic + Lomax simulation).
4. **Controlled — detect-and-isolate:** TE, post-isolation $g(\tau)$ and k, and epidemic simulations (naive vs. explicit).
5. **Controlled — gathering size restrictions:** Outbreak probability or effective k vs. gathering cap.
6. **Identifiability:** Posterior distributions for ψ; power curves vs. sample size and R_0.

1 Introduction

Individual-level variation can have a profound impact on an epidemic's trajectory. Interpersonal variation in infectiousness determines how likely an epidemic is to become established and how explosive it may be [Lloyd-Smith et al., 2005]. Likewise, individual-level stochastic effects early in an outbreak can durably shift the timing of an epidemic as a whole [Morris et al., 2024]. While the mean-field dynamics of an epidemic are dictated by two population-level quantities — the basic reproduction number (R_0) and the intrinsic generation interval distribution ($g(\tau)$) [Wallinga and Lipsitch, 2007] — stochastic departures from the mean depend on the individual-level heterogeneity from which these population-level quantities emerge.

Both R_0 and $g(\tau)$ are expectations of individual-level quantities: $R_0 = E_i[\nu_i]$, where ν_i is the individual reproduction number [Lloyd-Smith et al., 2005], and $g(\tau) = E_i[\xi_i(\tau)]$, where $\xi_i(\tau)$ is a probability density describing how person i's secondary infections are distributed in time. The parameters R_0 and $g(\tau)$ can be combined into a single object, $A(\tau) = R_0\, g(\tau)$, which captures both the total expected transmission and its distribution over time [Breda et al., 2012]. Analogously, the individual infectiousness profile or infectiousness kernel, $a_i(\tau) = \nu_i \xi_i(\tau)$, captures the expected number and timing of person i's secondary infections [Champredon and Dushoff, 2015, Svensson, 2007]. Substantial work has gone into understanding how variation in ν_i impacts epidemic dynamics; however, the impact of variation in $\xi_i(\tau)$ — independent of ν_i, R_0, and $g(\tau)$ — is less well developed.

Since $A(\tau)$ is an expectation over individuals, it need not resemble any single person's infectiousness profile. The SEIR model provides a clear illustration: it assumes each individual undergoes a latent period $x_i \sim \mathrm{Exp}(\gamma)$ during which they are not infectious, followed by an infectious period $y_i \sim \mathrm{Exp}(\alpha)$ during which they transmit at rate β. The resulting "stepwise" individual infectiousness profile is

$$a_i(\tau) = \begin{cases} \beta & \tau \in [x_i, x_i + y_i] \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

Averaging over individuals yields the population-level infectiousness profile

$$A_{\mathrm{SEIR}}(\tau) = E_i[a_i(\tau)] = \beta \frac{\gamma}{\gamma - \alpha}(e^{-\alpha\tau} - e^{-\gamma\tau}) \tag{2}$$

— a smooth, bell-shaped curve that emerges from a collection of discontinuous, rectangular $a_i(\tau)$ (**Figure XX**). Importantly, other choices of $a_i(\tau)$ can produce the same $A_{\mathrm{SEIR}}(\tau)$. Setting $a_i(\tau) \equiv A_{\mathrm{SEIR}}(\tau)$ for all individuals gives a "smooth" profile in which each person's infectiousness follows the population average. At the opposite extreme, $a_i(\tau) = (\beta/\alpha)\,\delta_{\tau_i^*}(\tau)$, with τ_i^* drawn from $g(\tau) = A_{\mathrm{SEIR}}(\tau)/R_0$, gives a "spike" profile in which each person's infectiousness is concentrated at a single moment. All three families — stepwise, smooth, and spike — produce the same $A(\tau)$ and hence the same deterministic, mean-field epidemic dynamics.

We have well-established methods for estimating R_0 and $g(\tau)$, but not for measuring $a_i(\tau)$. Doing so requires both accurate contact tracing and precise inference of infection times — data that are rarely available. Viral load kinetics offer an indirect window into the timing of infectiousness, but the mapping is far from straightforward: viral load, viable virus shedding, and infectiousness are distinct quantities. Nevertheless, we can reason about the likely shape of $a_i(\tau)$. The full variability in $g(\tau)$ arises from both within-individual and between-individual variation in infection timing. At one extreme, setting $a_i(\tau) \equiv A(\tau)$ attributes all variability to within-individual variation, but this is unlikely: individuals almost certainly differ in the

timing of their peak infectiousness, suggesting that $a_i(\tau)$ is generally narrower than $A(\tau)$. Indeed, some evidence suggests that the window of peak transmissibility for SARS-CoV-2 and influenza may be very short — less than a day [Goyal et al., 2021] — implying that much of the variation in $g(\tau)$ arises from between-individual differences in when that window occurs.

This raises a natural question: if measuring $a_i(\tau)$ is so challenging, is there any reason to try? Does it add anything beyond knowing R_0 and $g(\tau)$? There are several reasons to believe it might. At the individual level, knowing whether one's infectiousness is concentrated in a brief burst or spread over a longer period is important for guiding personal behavior. At the population level, a more punctuated $a_i(\tau)$ could meaningfully alter stochastic epidemic dynamics: individual-level stochastic effects early in an outbreak can cascade through the rest of the epidemic trajectory [Morris et al., 2024]. The shape of $a_i(\tau)$ also interacts with control measures that target individual infectiousness — such as symptom- or test-based screening and isolation — and with time-varying contact patterns. The impact of punctuated infectiousness has been examined as a sensitivity analysis in several epidemiological studies [Larremore et al., 2021] **Others?**, but a systematic understanding of its effect on epidemic dynamics is lacking.

In this paper, we ask: holding $A(\tau)$ constant, how do different choices of $a_i(\tau)$ impact epidemic dynamics? And how would we determine what $a_i(\tau)$ looks like in practice? We focus on the "punctuatedness" of $a_i(\tau)$ — the degree to which an individual's infectiousness is concentrated in time — as the key parameter of interest. This is equivalent to asking how the decomposition of generation interval variability into within-individual *vs.* between-individual components affects epidemic behavior. We restrict attention to pathogens with unimodal generation interval distributions and short transmission timescales (*e.g.*, respiratory viruses), though the framework extends in principle to pathogens with more complex dynamics. Interpersonal variation in the overall *level* of infectiousness has been studied extensively [Lloyd-Smith et al., 2005]; here, we focus specifically on interpersonal variation in the *timing* of infectiousness, holding all population-level parameters — including R_0 and $g(\tau)$ — constant. Our analyses therefore concern stochastic fluctuations around the mean-field solution. We also address the identifiability of the punctuatedness parameter and examine how inferential power changes as a function of R_0.

2 Modeling individual infectiousness

2.1 *The time-shifted Gamma model*

While our interest is in the general question of how $a_i(\tau)$, independent of $A(\tau)$, impacts epidemic dynamics, we introduce here a specific modeling framework — general enough to capture the essential features — that allows us to make precise claims. We seek a definition of $a_i(\tau)$ satisfying three requirements: (i) $A(\tau) = E_i[a_i(\tau)]$, as required by definition; (ii) $\nu_i = R_0$ for all i, so that there is no inherent interpersonal variation in total infectiousness; and (iii) the width of $a_i(\tau)$ is governed by a single smoothness parameter $\psi \in [0, 1]$, where $\psi = 1$ yields $a_i(\tau) \equiv A(\tau)$ (the smooth case, in which each person's individual infectiousness profile matches the population average) and $\psi \to 0$ makes $a_i(\tau)$ a delta function (the spike case, in which all of a person's secondary infections occur at a single moment).

We assume the intrinsic generation interval distribution $g(\tau)$ follows a Gamma(α, β) distribution with $\beta = \alpha/\bar{g}$, where $\bar{g}$ is the mean generation time — a common choice in applied epidemiology. Define each person's individual generation interval distribution as

$$\xi_i(\tau) = \begin{cases} 0 & \tau \leq l_i \\ f_\epsilon(\tau - l_i) & \tau > l_i \end{cases} \tag{3}$$

where $l_i \sim$ Gamma$((1-\psi)\alpha, \beta)$ is a random latent period and f_ϵ is the Gamma$(\psi\alpha, \beta)$ density. Setting $a_i(\tau) = R_0\, \xi_i(\tau)$ immediately satisfies requirement (ii). For requirement (iii), as $\psi \to 1$ the latent period vanishes ($l_i \to 0$) and f_ϵ approaches the full Gamma(α, β) density, so $\xi_i(\tau) = g(\tau)$ for all i; as $\psi \to 0$, f_ϵ concentrates to a point mass and $\xi_i(\tau)$ becomes a delta function at $\tau = l_i$.

For requirement (i), observe that the time from infection to a secondary infection attempt is $\tau = l_i + U_i$, where $U_i \sim$ Gamma$(\psi\alpha, \beta)$ is the draw from f_ϵ. Since l_i and U_i are independent Gamma random variables with a common rate β, their sum follows a Gamma$((1-\psi)\alpha + \psi\alpha, \beta) =$ Gamma(α, β) distribution by the additive property of the Gamma family. Taking expectations over individuals, $E_i[\xi_i(\tau)] = g(\tau)$, so $E_i[a_i(\tau)] = R_0\, g(\tau) = A(\tau)$ as required. Notably, this convergence does not require the population to share a common value of ψ: individual-specific values ψ_i drawn from any distribution on $[0, 1]$ still yield $E_i[a_i(\tau)] = A(\tau)$ (**Supplementary Methods**).

2.2 *Allowing for time-varying contacts*

The formulation above assumes time-homogeneous contacts, which is what allows $a_i(\tau)$ and $A(\tau)$ to depend only on time since infection τ and not on calendar time t. We follow this convention for most of our analyses. However, the interaction between punctuated infectiousness and time-varying contacts turns out to be consequential, so we briefly introduce the relevant generalization and two contact models that serve as test beds.

Under time-uniform contacts, $a_i(\tau) = R_0\, \xi_i(\tau)$, where R_0 and $\xi_i(\tau)$ are defined with respect to a reference contact process $\bar{c} \equiv 1$. To allow for time-varying contacts, we introduce a contact modulator $c_i(t)$, a nonnegative function describing person i's proportional deviation from the reference contact level at calendar time t. We require $E_t[c_i(t)] = 1$, so that R_0 retains its interpretation as the expected number of secondary infections under average contact conditions; this is satisfied by any periodic function with mean 1 or any

stationary random process with mean 1. The individual infectiousness profile generalizes to

$$a_i(t_i, \tau) = R_0\, \xi_i(\tau)\, c_i(t_i + \tau) \tag{4}$$

where t_i is person i's infection time. This expression makes explicit that, when contacts vary, a person's realized infectiousness depends not only on their biological profile $\xi_i(\tau)$ but also on when they happen to be infectious. From Eq. 4, we can define the realized individual reproduction number,

$$\nu_i^{\mathrm{rel}}(t_i) = R_0 \int_0^\infty \xi_i(\tau)\, c_i(t_i + \tau)\, d\tau \tag{5}$$

and the realized individual generation interval distribution,

$$\xi_i^{\mathrm{rel}}(t_i, \tau) = \frac{\xi_i(\tau)\, c_i(t_i + \tau)}{\displaystyle\int_0^\infty \xi_i(\tau')\, c_i(t_i + \tau')\, d\tau'} \tag{6}$$

so that $a_i(t_i, \tau) = \nu_i^{\mathrm{rel}}(t_i)\, \xi_i^{\mathrm{rel}}(t_i, \tau)$. When contacts are time-uniform ($c_i \equiv 1$), these reduce to $\nu_i^{\mathrm{rel}} = R_0$ and $\xi_i^{\mathrm{rel}} = \xi_i$. When contacts vary, ν_i^{rel} fluctuates around R_0: a person whose brief window of infectiousness coincides with a period of high contact activity will have $\nu_i^{\mathrm{rel}} > R_0$, while one whose window falls during a low-contact period will have $\nu_i^{\mathrm{rel}} < R_0$. Crucially, this effect is stronger when $\xi_i(\tau)$ is narrow (small ψ), because a concentrated profile effectively samples the contact process at a single point, whereas a broad profile averages over it.

As test cases, we consider two models. For the *periodic* contact model, we set

$$c_i(t) = 1 - A\cos(\omega t) \quad \forall\, i \tag{7}$$

with amplitude $A \in [0, 1]$ and angular frequency $\omega > 0$. The default $\omega = 2\pi\,\mathrm{day}^{-1}$ models diurnal contact variation, with higher contact rates during the day and lower rates at night. Setting $A = 0$ recovers the reference process. In this model, all individuals share the same contact trajectory, so there is temporal variation within each person's contacts but no variation across individuals.

For the *Lomax* contact model, each person's contact modulator is a piecewise-constant process that switches to a new level at the points of a Poisson process with rate λ, so that sojourn times at each level are exponentially distributed with mean $1/\lambda$. The levels are drawn independently from a Lomax (Pareto Type II) distribution with shape α_P and scale σ,

$$f_{c^*}(c) = \frac{\alpha_P}{\sigma}\left(1 + \frac{c}{\sigma}\right)^{-(\alpha_P+1)}, \qquad c \geq 0 \tag{8}$$

with $\alpha_P > 2$. The Lomax distribution has mean $\sigma/(\alpha_P - 1)$, so requiring a unit mean pins down $\sigma = \alpha_P - 1$, leaving a one-parameter family indexed by the tail exponent α_P. Smaller α_P yields a heavier tail and rarer but more extreme contact levels; larger α_P concentrates the distribution around 1. In contrast to the periodic model, each person's contact trajectory is unique, but the population average is constant: $C(t) = E_i[c_i(t)] = 1$.

2.3 *Simulation approach*

We investigate the impact of punctuatedness on epidemic dynamics using both closed-form analytic results (derived in the **Supplementary Methods**) and stochastic simulations. Stochastic simulations use a branching-process-like algorithm in a finite population of $N = 10{,}000$ susceptible individuals with a single initial infectious individual. For each infected person i:

1. A latent period l_i is drawn from $\text{Gamma}((1-\psi)\alpha, \beta)$, defining the individual generation interval distribution $\xi_i(\tau)$ via Eq. 3. When contacts are non-uniform, a contact trajectory $c_i(t)$ is also drawn from the relevant contact model.
2. The number of secondary infection attempts is drawn as $X_i \sim \text{Poisson}(R_0)$.
3. The timing of each attempt relative to infection is drawn from $\xi_i(\tau)$.
4. When the contact process is non-uniform ($c_i \neq 1$), Poisson thinning is applied: each attempt at calendar time $t_i + \tau$ is accepted with probability proportional to $c_i(t_i + \tau)$, ensuring that the realized infection rate matches $a_i(t_i, \tau)$.
5. Each accepted attempt is paired with a uniformly chosen individual from the population: if the recipient is susceptible, infection occurs; otherwise, the attempt fails.

Full details are provided in the **Supplementary Methods**. For illustration, we focus on three sets of epidemiological parameters roughly pertaining to influenza-like, SARS-CoV-2 omicron-like, and measles-like pathogens (**Table XX**).

3 The impact of punctuated infectiousness profiles on uncontrolled epidemics

3.1 Punctuated individual infectiousness profiles generate later and more variably-timed epidemics.

Early-outbreak stochasticity can impact when an epidemic becomes established and begins its exponential growth phase [Morris et al., 2024]. This time shift durably impacts the epidemic's full trajectory. The effect of this early-outbreak stochasticity can be modeled as a random time shift applied to the deterministic epidemic trajectory [Morris et al., 2024]. Here, we demonstrate that the distribution of this time shift — both its mode and variance — depend on ψ.

Let contacts be uniform over time and across individuals ($c_i(t) \equiv 1 \, \forall \, i$), and let $b_i(\tau)$ follow the time-shifted Gamma model (Eq. 3). Drawing upon branching process theory, the cumulative incidence over time, $Z(t)$, during the exponential growth phase can be approximated as a deterministic exponential function with a random initial scaling factor W:

$$Z(t) \approx W \cdot e^{rt} \tag{9}$$

Here, r is the early-outbreak exponential growth rate and $E[W]$ is the initial number of infectious individuals, I_0. For clarity of the derivation, we use $E[W] = I_0 = 1$ throughout.

The random scaling factor W can, through a change of variables, be expressed as a random time shift:

$$Z(t) \approx e^{r(t+\varsigma)} \tag{10}$$

where

$$\varsigma = \frac{\log W - \log E[W]}{r} = \frac{\log W}{r} \tag{11}$$

when $E[W] = 1$.

Given this setup, it can be shown (**Supplementary Methods**) that

$$\mathrm{Var}[W] = \frac{\left(\frac{1+z}{1+2z}\right)^{\alpha} + \left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha} - 1}{1 - \left(\frac{1+z}{1+2z}\right)^{\alpha}} \tag{12}$$

where $z = R_0^{1/\alpha} - 1$. The punctuation parameter ψ enters the expression only once, in the middle term of the numerator; thus, $\mathrm{Var}[W]$ and ψ are monotonically related, where narrower individual infectiousness profiles ($\psi \to 0$) maximize the variance and broader profiles ($\psi \to 1$) minimize it.

We can also derive an approximate expression for the mode of the distribution of ς, $f_\varsigma(\tau)$. To approximate $f_\varsigma(\tau)$, we can use a two-moment version of the moment matching approach of Morris et al. [2024]. These moments are

$$E[W] = 1, \qquad E[W^2] = \frac{\left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha}}{1 - \left(\frac{1+z}{1+2z}\right)^{\alpha}} \tag{13}$$

where the first follows from the property $E[W] = I_0$ and the second follows from the variance calculation derived above, with $E[W^2] = \mathrm{Var}[W] + (E[W])^2 = \mathrm{Var}[W] + 1$. With these moments in hand it can be

shown (**Supplementary Methods**) that

$$\mathrm{Mode}[\varsigma] = \frac{1}{r}\log\left(\frac{2\left[1-\left(\frac{1+z}{1+2z}\right)^{\alpha}\right] - p\left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha}}{p\left[1-\left(\frac{1+z}{1+2z}\right)^{\alpha}\right]}\right) \tag{14}$$

provided the numerator within the logarithm is positive; when the numerator is zero or negative, no finite modal time shift exists and $\mathrm{Mode}[\varsigma] \to -\infty$. Again, ψ enters the expression only once, with narrow individual infectiousness profiles ($\psi \to 0$) minimizing the mode (making it more negative, *i.e.*, causing the typical epidemic to be shifted later) and broad individual infectiousness profiles ($\psi \to 1$) maximizing it (causing the typical epidemic to be shifted earlier).

Note that the ψ-dependence of both $\mathrm{Var}[W]$ and $\mathrm{Mode}[\varsigma]$ enter through the expression

$$\left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha} \tag{15}$$

This base is monotonically increasing in R_0 (**Supplementary Methods**), with $R_0 = 1$ implying $z = 0$, causing the ψ-dependence to disappear. Thus, increasing R_0 amplifies the ψ-dependency of $\mathrm{Var}[W]$ and $\mathrm{Mode}[\varsigma]$.

[A comment on intuition behind these results: narrower infectiousness profiles make it so that the offspring all inherit variation in the latent period(?), inflating the overall variance of the process. In a few epidemics, we have explosive spread early on, causing really fast take-offs; but often we also miss the early transmissions that trickle in causing lots of late take-offs. For the mode, we have a minimum of order statistics story, where the earliest offspring tends to come later the more punctuated the profile is.]

Stochastic simulations confirm the analytic results. The time to 100 infections varies with ψ, especially for pathogens with high R_0 (SARS-CoV-2 omicron, measles). For measles, by 28 days, only xx% of simulated epidemics reached 100 infections by day 28 for the spike case, while xx% reached the same threshold for the smooth case. Thus, punctuation of the individual infectiousness profile can substantially impact the timing of epidemic establishment, especially when R_0 is high.

3.2 Narrow individual infectiousness profiles yield greater uncertainty in epidemic growth rate estimates

Punctuated individual infectiousness profiles also make empirical estimates of the early-epidemic exponential growth rate r more uncertain (**Figure XX**).

3.3 Narrow biological infectiousness profiles interact with periodic contact patterns to generate overdispersion in secondary infections

When contacts are uniform in time, the "punctuatedness" of biological infectiousness (ψ) has no effect on the secondary infection distribution: every individual produces a Poisson(R_0)-distributed number of infections, and there is no overdispersion ($k \to \infty$) irrespective of ψ. However, if contacts vary over time, narrower biological profiles "see" more of this contact variation, generating overdispersion in the secondary infection distribution. We illustrate this using the periodic contact model, $c_i(t) = 1 - A\cos(\omega t + \Phi)$, for

which a closed-form expression for k can be obtained (**Supplementary Methods**):

$$k = \frac{2}{A^2}\left(1 + \frac{\omega^2}{\beta^2}\right)^{\psi\alpha} \tag{16}$$

where A is the amplitude of the contact oscillation and ω is its angular frequency. Here, the prefactor $2/A^2$ sets the overall overdispersion scale, and the bracketed factor quantifies the failure of the biological profile's temporal window to average out the contact oscillation, with $(1 + \omega^2/\beta^2)^{-\psi\alpha}$ being exactly the squared characteristic function of the Gamma$(\psi\alpha, \beta)$ density at the contact frequency ω. Since this factor is strictly increasing in ψ for any $\omega > 0$, narrower biological profiles always produce more overdispersion in the presence of periodic contact variation. The effect disappears in the spike limit only if $\omega \to 0$ or $A \to 0$ (no temporal contact variation), and grows arbitrarily large in the smooth limit if ω/β is large (contact oscillations much faster than the profile window, so the profile averages them out). Simulations extend this result to heavier-tailed, non-periodic contact heterogeneity, for which the same mechanism holds qualitatively (**Figure XX**).

4 The impact of punctuated individual infectiousness on detect-and-isolate interventions

The impact of a detect-and-isolate protocol can be measured in terms of its "test efficacy" (TE) [Middleton and Larremore, 2024]. The TE measures the amount of infectiousness that a detection protocol (possibly test-based, but this extends also to symptom-based isolation and other general isolation triggers) prevents.

When detection is unrelated to the timing of a person's infectiousness, the individual infectiousness profile cancels out, and the generation interval distribution is all that is neede to calculate TE. In the realistic scenario that detectability is anchored to the person's infectiousness somehow —*e.g.*, symptoms arise at a time related to peak infectiousness, or detectability can occur during some window that extends before and after a person's peak infectiousness – the shape of the individual infectiousness profile impacts TE. In general, when detection tends to occur before peak infectiousness, TE is higher for narrower infectiousness profiles; when detection usually happens after peak infectiousness, TE is lower for narrow infectiousness profiles (**Figure XX**).

TE, however, only reflects part of the story with individual infectiousness. Detect-and-isolate protocols have two additional impacts whose strength depends on the shape of the individual infectiousness profile: they re-shape the overall generation interval distribution by cutting off late transmissions (this happens moreso for smooth profiles than for spiky ones), and they can increase overdispersion in the secondary infection distribution (this happens moreso for smooth profiles). As a result, the naive simulations of epidemics using $R_{\text{eff}} = R_0(1 - \text{TE})$ fail to capture key elements of epidemics that result when the detect-and-isolate protocol is explicitly modeled and the individual infectiousness profile is accounted for.

We illustrate this using two specific detect-and-isolate protocols: one in which symptoms arise with some lag μ relative to peak infectiousness and variance σ^2, and one where there exists a detectability window around the peak infectiousness and people participate in regular testing.

Results for g, k, and TE for symptom-based screening.

Results for g, k, and TE for test-based screening.

Epidemic simulations illustrating differences.

5 The impact of individual infectiousness on gathering size restrictions

6 Identifiability and inference of the individual infectiousness profile's shape

While the concentration of the individual infectiousness profile can impact how an epidemic unfolds, we lack frameworks for estimating it. Many approaches exist for estimating the generation interval, but there has not been a systematic approach developed to assess how individual infectiousness varies and to specify where on the continuum we sit—i.e., how much of the variation in the generation interval distribution is attributable to variation between individuals, and how much to variation in infection timing within individuals.

We begin by assessing under which conditions we can hope to infer the shape of the individual infectiousness profile. We restrict to gamma-distributed timings, and consider cases where we assume that ψ is shared across individuals and where ψ comes from a beta distribution governed by hyperparameters α and β. Then, when ψ is identifiable, we introduce a hierarchical Bayesian inference strategy to infer the shape of the individual infectiousness profile given contact tracing data.

For this section: let's imagine that there's a single *psi* value shaed among everyone. We can use a Beta prior. Two things to consider: how do we estimate ψ given a list of infection times (assume we know them exactly)? And how do we estimate ψ given infection times and knowledge of $c_i(t)$? My hope is that this will emphasize that knowing the "negatives" – i.e., the contacts that "should" have led to transmission but did not – improves our inferential power substantially. Then, I want to think about how to achieve a given power in each scenario across R0, assuming some fixed generation interval distribution. That should be enough for now. We should probably include some amount of uncertainty in infection times for this base analysis, and we should post code for running the analysis given a line list of infection times and contacts. In the Discussion, we need to address the fact that there could be different ψ_i values across individuals, themselves drawn from a distribution, so we'd want some kind of hierarchical Bayesian model for inference here; also, we don't consider differences in the height (i.e. inherent biological differences in ν_i / b_i), which is another key thing to include in the inference.

7 Discussion

Uncertain early exponential growth rate calculations feed into estimates of R_0 — they are a critical ingredient for most R_0 estimation formulae.

The superspreading mechanism introduced here is novel, and implies that some amount of superspreading cannot be traced *a priori* to the individual: there is no such thing as identifying superspreaders, since the superspreader in this case may not have any more inherent infectiousness or a greater number of contacts; instead, they were just infected at the wrong time.

Accounting for the individual infectiousness profile is critical for anticipating the imapct of outbreak interventions. Simply calculating reductions in infectiousness is not enough: we get differences in overdispersion and the generation interval distribution that are sensitive to the punctuatedness of $a_i(\tau)$ and can lead to profoundly different epidemic dynamcis than a naive approach of simulating an equivalent epidemic without changing overdispersion or the generation interval distribution.

Estiamting the shape of the individual infectiousness profile can be challenging given the need for precise estimates of time of infection and a good balance of depth and breadth of sampling. Fortunately, identifiability of the shape of $a_i(\tau)$ is better in situations where it matters most, *i.e.*, when R_0 is high.

Limitations.

Concluding thoughts.

References

D. Breda, O. Diekmann, W. F. De Graaf, A. Pugliese, and R. Vermiglio. On the formulation of epidemic models (an appraisal of Kermack and McKendrick). *Journal of Biological Dynamics*, 6(sup2):103–117, September 2012. ISSN 1751-3758, 1751-3766. doi: 10.1080/17513758.2012.716454. URL http://www.tandfonline.com/doi/abs/10.1080/17513758.2012.716454.

David Champredon and Jonathan Dushoff. Intrinsic and realized generation intervals in infectious-disease transmission. *Proceedings of the Royal Society B: Biological Sciences*, 282(1821):20152026, December 2015. ISSN 0962-8452, 1471-2954. doi: 10.1098/rspb.2015.2026. URL https://royalsocietypublishing.org/doi/10.1098/rspb.2015.2026.

Ashish Goyal, Daniel B Reeves, E Fabian Cardozo-Ojeda, Joshua T Schiffer, and Bryan T Mayer. Viral load and contact heterogeneity predict SARS-CoV-2 transmission and super-spreading events. *eLife*, 10: e63537, February 2021. ISSN 2050-084X. doi: 10.7554/eLife.63537. URL https://elifesciences.org/articles/63537.

William Ogilvy Kermack and A. G. McKendrick. A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character*, 115(772):700–721, August 1927. ISSN 0950-1207, 2053-9150. doi: 10.1098/rspa.1927.0118. URL https://royalsocietypublishing.org/rspa/article/115/772/700/2165/A-contribution-to-the-mathematical-theory-of.

Daniel B. Larremore, Bryan Wilder, Evan Lester, Soraya Shehata, James M. Burke, James A. Hay, Milind Tambe, Michael J. Mina, and Roy Parker. Test sensitivity is secondary to frequency and turnaround time for COVID-19 screening. *Science Advances*, 7(1):eabd5393, January 2021. ISSN 2375-2548. doi: 10.1126/sciadv.abd5393. URL https://www.science.org/doi/10.1126/sciadv.abd5393.

J. O. Lloyd-Smith, S. J. Schreiber, P. E. Kopp, and W. M. Getz. Superspreading and the effect of individual variation on disease emergence. *Nature*, 438(7066):355–359, November 2005. ISSN 0028-0836, 1476-4687. doi: 10.1038/nature04153. URL https://www.nature.com/articles/nature04153.

Casey Middleton and Daniel B. Larremore. Modeling the transmission mitigation impact of testing for infectious diseases. *Science Advances*, 10(24):eadk5108, June 2024. ISSN 2375-2548. doi: 10.1126/sciadv.adk5108. URL https://www.science.org/doi/10.1126/sciadv.adk5108.

Dylan Morris, John Maclean, and Andrew J. Black. Computation of random time-shift distributions for stochastic population models. *Journal of Mathematical Biology*, 89(3):33, September 2024. ISSN 0303-6812, 1432-1416. doi: 10.1007/s00285-024-02132-6. URL https://link.springer.com/10.1007/s00285-024-02132-6.

Åke Svensson. A note on generation times in epidemic models. *Mathematical Biosciences*, 208(1):300–311, July 2007. ISSN 00255564. doi: 10.1016/j.mbs.2006.10.010. URL https://linkinghub.elsevier.com/retrieve/pii/S0025556406002094.

J Wallinga and M Lipsitch. How generation intervals shape the relationship between growth rates and reproductive numbers. *Proceedings of the Royal Society B: Biological Sciences*, 274(1609):599–604, February 2007. ISSN 0962-8452, 1471-2954. doi: 10.1098/rspb.2006.3754. URL https://royalsocietypublishing.org/doi/10.1098/rspb.2006.3754.

Acknowledgements

Funding

Author contributions

Supplementary Information

Supplementary Methods

Parameter structure

The punctuated Gamma model has four epidemiological parameters: the basic reproduction number R_0, the early-epidemic growth rate r, and the two parameters of the generation interval distribution, α (shape) and β (rate), which together determine the mean generation time $\bar{g} = \alpha/\beta$. These four parameters are not all free: the Euler–Lotka equation

$$1 = R_0 \int_{\tau=0}^{\infty} e^{-r\tau} \frac{\beta^\alpha}{\Gamma(\alpha)} \tau^{\alpha-1} e^{-\beta\tau} d\tau = R_0 \left(\frac{\beta}{\beta + r} \right)^\alpha \tag{S1}$$

provides one constraint among them, leaving three free parameters. We choose to work with $(R_0, \bar{g}, \alpha)$ as the free parameters, since each has a direct epidemiological interpretation:

- R_0: the basic reproduction number, measuring transmissibility;
- $\bar{g}$: the mean generation time;
- α: the shape parameter of the generation interval distribution, where larger α corresponds to a more concentrated (less variable) generation interval distribution for fixed $\bar{g}$.

Given $(R_0, \bar{g}, \alpha)$, the remaining parameters are fully determined. The rate parameter β follows immediately from the definition of the mean:

$$\beta = \frac{\alpha}{\bar{g}}, \tag{S2}$$

and the early-epidemic growth rate r follows from the Euler–Lotka equation:

$$r = \beta \left(R_0^{1/\alpha} - 1 \right) = \frac{\alpha}{\bar{g}} \left(R_0^{1/\alpha} - 1 \right). \tag{S3}$$

The punctuation parameter $\psi \in [0, 1]$ is the only remaining free parameter, and it is the sole quantity that is not identifiable from population-level epidemic data: since $A(\tau)$ is invariant across ψ by construction, the parameters $(R_0, \bar{g}, \alpha)$ can in principle be estimated from the observed epidemic trajectory independently of ψ. The consequences of varying ψ while holding $(R_0, \bar{g}, \alpha)$ fixed are therefore attributable purely to changes in the individual-level temporal structure of infectiousness, with no confounding from changes in the population-level transmission kernel.

Derivation of Var[W]

For a branching process $Z(t)$ conditional on non-extinction, $e^{-rt}Z(t) \to W$ almost surely as $t \to \infty$. Here, r is the early-epidemic exponential growth rate and W is a random variable that effectively scales the branching process' initial condition, capturing stochastic fluctuations in the early-epidemic process.

With a change of variables, we can translate W into a random initial time shift:

$$\varsigma = \frac{\log W - \log E[W]}{r} = \frac{\log W}{r} \tag{S4}$$

when $I_0 = E[W] = 1$. Our goal is to understand how $\mathrm{Var}[\varsigma]$ depends on the punctuation parameter ψ. Since ς and W are monotonically related, we can derive an expression for $\mathrm{Var}[W]$ and will interpret its implications for $\mathrm{Var}[\varsigma]$ at the end.

To start, we consider an index case i infected at time $t = 0$ who produces $\chi_i \sim \mathrm{Poisson}(R_0)$ offspring. Under the Gamma time-shift model, the offspring j from this index case are infected at times $\tau_j = l_i + \epsilon_j$, where l_i is person i's latent period and ϵ_j is the additional lag between l_i and person j's infection time, with

$$l_i \sim \mathrm{Gamma}((1-\psi)\alpha, \beta), \qquad \epsilon_j \sim \mathrm{Gamma}(\psi\alpha, \beta)$$

This ensures that the population-level generation interval distribution is

$$g(\tau) \sim \mathrm{Gamma}(\alpha, \beta)$$

We can define some useful quantities that will arise later in the derivation. First, from the Euler-Lotka equation under $g(\tau)$ [Wallinga and Lipsitch, 2007]:,

$$1 = R_0 \int_{\tau=0}^{\infty} e^{-r\tau} \frac{\beta^\alpha}{\Gamma(\alpha)} \tau^{\alpha-1} e^{-\beta\tau} d\tau$$

which simplifies to

$$1 = R_0 \Big(\frac{\beta}{\beta + r}\Big)^\alpha = R_0 \Big(\frac{1}{1+z}\Big)^\alpha$$

where $z = r/\beta = r\bar{g}/\alpha = R_0^{1/\alpha} - 1$, a dimensionless quantity that simplifies expressions and allows one to easily plug in different combinations of epidemiological parameters. Here, $\bar{g}$ is the mean of the generation interval distribution $g(\tau)$.

Next, we define

$$\rho_k = (E[e^{-kr\tau}])^{1/\alpha} = \frac{\beta}{\beta + kr} = \frac{1}{1+kz}$$

This allows us to re-write the Euler-Lotka equation:

$$1 = R_0 \rho_1^\alpha$$

Next, we introduce the distributional fixed point equation, which is a standard starting point for computing moments of W. This equation exploits the self-similarity of the branching process, in which an epidemic initiated by a single individual is, in distribution, a scaled and time-shifted copy of the whole epidemic. At large t, the epidemic seeded by the index case i consists of the downstream epidemics caused

by their offspring j, which have grown approximately to size $Z_j(t) = W_j e^{r(t-\tau_j)}$. That is,

$$Z(t) \approx \sum_{j=1}^{\chi_i} W_j e^{rt} e^{-r\tau_j}$$

Multiplying through by e^{-rt}:

$$W \stackrel{d}{=} \sum_{j=1}^{\chi_i} e^{-r\tau_j} W_j$$

We now use this expression to calculate $\mathrm{Var}[W]$. Since randomness enters through both W and τ, we first condition on τ_j using the law of total variance:

$$\mathrm{Var}[W] = E[\mathrm{Var}\Big[\sum_j e^{-r\tau_j} W_j | \tau_j\Big]] + \mathrm{Var}[E\Big[\sum_j e^{-r\tau_j} W_j | \tau_j\Big]]$$

$$= E\Big[\sum_j e^{-2r\tau_j} \cdot \mathrm{Var}[W]\Big] + \mathrm{Var}\Big[\sum_j e^{-r\tau_j} \cdot E[W]\Big]$$

$$= E\Big[\sum_j e^{-2r\tau_j}\Big] \cdot \mathrm{Var}[W] + (E[W])^2 \cdot \mathrm{Var}\Big[\sum_j e^{-r\tau_j}\Big]$$

Since $E[W] = I_0 = 1$, we have

$$\mathrm{Var}[W] = \mathrm{Var}[W] E\Big[\sum_j e^{-2r\tau_j}\Big] + \mathrm{Var}\Big[\sum_j e^{-r\tau_j}\Big]$$

Solving for $\mathrm{Var}[W]$,

$$\mathrm{Var}[W] = \frac{\mathrm{Var}[\sum_j e^{-r\tau_j}]}{1 - E[\sum_j e^{-2r\tau_j}]}$$

To simplify the denominator, note that

$$E\Big[\sum_{j=1}^{\chi_i} e^{-2r\tau_j}\Big] = E[\chi_i] E[e^{-2r\tau_j}] = R_0 E[e^{-2r\tau_j}] = R_0 \rho_2^{\alpha}$$

Thus, the variance expression simplifies to

$$\mathrm{Var}[W] = \frac{\mathrm{Var}[\sum_j e^{-r\tau_j}]}{1 - R_0 \rho_2^{\alpha}}$$

The denominator does not depend on ψ, so we proceed by examining the numerator:

$$\mathrm{Var}[\sum_j e^{-r\tau_j}] = E[(\sum_j e^{-r\tau_j})^2] - E[\sum_j e^{-r\tau_j}]^2 = E[(\sum_j e^{-r\tau_j})^2] - 1$$

Then,

$$E[(\sum_j e^{-r\tau_j})^2] = E[\sum_j e^{-2r\tau_j}] + E[\sum_{j \neq k} e^{-r(\tau_j + \tau_k)}]$$

$$= R_0 \rho_2^{\alpha} + E[\chi_i(\chi_i - 1)] E[e^{-r\tau_j} e^{-r\tau_k}]$$

$$= R_0\rho_2^\alpha + R_0^2 E[e^{-r\tau_j}e^{-r\tau_k}]$$

Here, the $\chi_i(\chi_i - 1)$ term enters because there are this many terms in the sum over $j \neq k$. Also,

$$E[\chi_i(\chi_i - 1)] = E[\chi_i^2] - E[\chi_i] = Var[\chi_i] + E[\chi_i]^2 - E[\chi_i] = R_0 + R_0^2 - R_0 = R_0^2$$

under the assumption that $\chi_i \sim \text{Poisson}(R_0)$.

Since $\tau_j = l_i + \epsilon_j$, and since l_i, ϵ_j, and ϵ_k are independent,

$$E[e^{-r\tau_j}e^{-r\tau_k}] = E[e^{-2rl_i}e^{-r\epsilon_j}e^{-r\epsilon_k}] = E[e^{-2rl_i}](E[e^{-r\epsilon}])^2 = \rho_2^{(1-\psi)\alpha} \cdot \rho_1^{2\psi\alpha}$$

Finally, we can assemble $\text{Var}[W]$:

$$\text{Var}[\sum_j e^{-r\tau_j}] = R_0\rho_2^\alpha + R_0^2\rho_2^{(1-\psi)\alpha}\rho_1^{2\psi\alpha} - 1$$

and thus, substituting in the z expressions for ρ_1 and ρ_2, we obtain

$$\text{Var}[W] = \frac{\left(\frac{1+z}{1+2z}\right)^\alpha + \left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha} - 1}{1 - \left(\frac{1+z}{1+2z}\right)^\alpha} \tag{S5}$$

which is Eq. 12.

Mode of the two-moment Gamma approximation for the time-shift distribution

Following a simplified version of the moment-matching approach of Morris et al. [2024], we approximate the distribution of W conditional on survival by a Gamma distribution matched to its first two moments. The unconditional moments of W are:

$$E[W] = 1 \tag{S6}$$

$$E[W^2] = \mathrm{Var}[W] + E[W]^2 = \frac{\left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha}}{1 - \left(\frac{1+z}{1+2z}\right)^{\alpha}} \tag{S7}$$

Conditioning on survival with probability p:

$$E[W \mid \mathrm{surv}] = \frac{E[W]}{p} = \frac{1}{p} \tag{S8}$$

$$E[W^2 \mid \mathrm{surv}] = \frac{E[W^2]}{p} = \frac{\left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha}}{p\left(1 - \left(\frac{1+z}{1+2z}\right)^{\alpha}\right)} \tag{S9}$$

where the division by p arises because

$$E[W^n] = E[W^n|\mathrm{surv}]P(\mathrm{surv}) + E[W^n|\mathrm{extinct}]P(\mathrm{extinct}) = E[W^n|\mathrm{surv}] \cdot p \tag{S10}$$

since $W = 0$ when the epidemic fails to establish. Thus, $E[W^n|\mathrm{surv}] = E[W^n]/p$.

Our goal is to approximate the distribution of W with a Gamma(k, λ) distribution, and specifically to estimate the mode of the distribution of $W|\mathrm{surv}$ with the mode of the moment-matched Gamma(k, λ) distribution. Note that the mode of a generic Gamma(k, λ) distribution is $(k-1)/\lambda$, and that the distribution has a mean of k/λ and a squared coefficient of variation of (CV^2) $1/k$. Thus, the parameters of the moment-matched Gamma distribution are

$$k = \frac{1}{\mathrm{CV}^2[W|\mathrm{surv}]} \tag{S11}$$

$$\lambda = \frac{k}{E[W|\mathrm{surv}]} = kp \tag{S12}$$

and the mode of the moment-matched distribution is

$$\mathrm{mode}[W|\mathrm{surv}] \approx \frac{k-1}{\lambda} = \frac{k-1}{kp} = \frac{1-1/k}{p} = \frac{1-\mathrm{CV}^2[W|\mathrm{surv}]}{p} \tag{S13}$$

The expression for CV^2 is

$$\mathrm{CV}^2(W|\mathrm{surv}) = \frac{\mathrm{Var}(W|\mathrm{surv})}{E[W|\mathrm{surv}]^2} = \frac{E[W^2|\mathrm{surv}] - E[W|\mathrm{surv}]^2}{E[W|\mathrm{surv}]^2} \tag{S14}$$

$$= \frac{p\left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha}}{1 - \left(\frac{1+z}{1+2z}\right)^{\alpha}} - 1 \tag{S15}$$

This moment-matched approximation is valid when $k \geq 1$ (*i.e.*, $\mathrm{CV}^2[W|\mathrm{surv}] \leq 1$). When $k < 1$, the approximating density is monotonically decreasing on $(0, \infty)$ with no interior mode.

The time-shift $\varsigma = \log(W)/r$ is a monotone transformation of W, so its modal value is obtained by transforming the mode of the Gamma approximation. Therefore:

$$\mathrm{Mode}[\varsigma] = \frac{1}{r} \log\left(\frac{1-V}{p}\right) \tag{S16}$$

Substituting V from Eq. (S15) and expressing in terms of $z = R_0^{1/\alpha} - 1$, this yields Eq. (14).

Monotonicity in R_0

Differentiating Eq. (S22) with respect to R_0, holding α fixed, and using $\frac{d}{dR_0}(2R_0^{1/\alpha}) = (2/\alpha)R_0^{1/\alpha-1}$:

$$\frac{\partial(\alpha \log \sigma)}{\partial R_0} = \frac{2}{R_0} - \frac{2R_0^{1/\alpha-1}}{2R_0^{1/\alpha}-1} = \frac{2}{R_0} \cdot \frac{R_0^{1/\alpha}-1}{2R_0^{1/\alpha}-1} = \frac{2(\mu-1)}{R_0(2\mu-1)} \tag{S17}$$

This is strictly positive for all $R_0 > 1$, since $\mu = R_0^{1/\alpha} > 1$ makes both the numerator $\mu - 1$ and denominator $2\mu - 1$ positive. The derivative vanishes at $R_0 = 1$ ($\mu = 1$), confirming that at the epidemic threshold there is no ψ-dependence in either $\mathrm{Var}(W)$ or $\mathrm{Mode}[\varsigma]$.

Monotonicity in α

Differentiating Eq. (S22) with respect to α, holding R_0 fixed. Write $g(\alpha) = \alpha \log(2R_0^{1/\alpha} - 1)$, so that $\alpha \log \sigma = 2 \log R_0 - g(\alpha)$ and it suffices to show $\partial g / \partial \alpha > 0$. By the product rule, using $\frac{\partial}{\partial \alpha} R_0^{1/\alpha} = -R_0^{1/\alpha} \log R_0 / \alpha^2$:

$$\frac{\partial g}{\partial \alpha} = \log(2\mu - 1) - \frac{2\mu \log \mu}{2\mu - 1} \tag{S18}$$

where we have substituted $\mu = R_0^{1/\alpha}$ and $\log R_0 = \alpha \log \mu$. To establish the sign, define $h(\mu) = (2\mu - 1) \log(2\mu - 1) - 2\mu \log \mu$, so that the condition $\partial g / \partial \alpha > 0$ is equivalent to $h(\mu) > 0$. Differentiating:

$$h'(\mu) = 2 \log(2\mu - 1) + 2 - 2 \log \mu - 2 = 2 \log\left(2 - \frac{1}{\mu}\right) \tag{S19}$$

For $\mu > 1$, we have $2 - 1/\mu > 1$, so $h'(\mu) > 0$. Since $h(1) = 0$ and $h'(\mu) > 0$ for all $\mu > 1$, it follows that $h(\mu) > 0$ for all $\mu > 1$, giving $\partial g / \partial \alpha > 0$ and hence:

$$\frac{\partial(\alpha \log \sigma)}{\partial \alpha} < 0 \tag{S20}$$

So $\alpha \log \sigma$ is strictly decreasing in α: more concentrated generation intervals (larger α, holding R_0 fixed) reduce the impact of punctuation on both $\mathrm{Var}(W)$ and $\mathrm{Mode}[\varsigma]$.

The modulating role of $\alpha \log \sigma$

The ψ-dependent term in both $\mathrm{Var}(W)$ and $\mathrm{Mode}[\varsigma]$ is $\sigma^{(1-\psi)\alpha}$, which can be written as $e^{(1-\psi)\alpha \log \sigma}$. The quantity $\alpha \log \sigma$ therefore acts as the fundamental modulator of ψ-dependence: larger $\alpha \log \sigma$ means the base $\sigma > 1$ is raised to a larger effective power, amplifying the spread in $\mathrm{Var}(W)$ and $\mathrm{Mode}[\varsigma]$ between the spike ($\psi \to 0$) and smooth ($\psi \to 1$) limits. We can express this quantity exactly in terms of R_0 and α alone. Starting from $\sigma = \mu^2/(2\mu - 1)$ with $\mu = R_0^{1/\alpha}$:

$$\log \sigma = 2 \log \mu - \log(2\mu - 1) = \frac{2 \log R_0}{\alpha} - \log(2R_0^{1/\alpha} - 1) \tag{S21}$$

and multiplying through by α:

$$\alpha \log \sigma = 2 \log R_0 - \alpha \log(2R_0^{1/\alpha} - 1). \tag{S22}$$

We now show that this expression is strictly increasing in R_0 and strictly decreasing in α, confirming the qualitative picture that punctuation ($\psi < 1$) matters more when R_0 is large and the generation interval is dispersed (small α).

Summary

The two partial derivatives establish that the modulating quantity $\alpha \log \sigma$ — and hence the sensitivity of $\mathrm{Var}(W)$ and $\mathrm{Mode}[\varsigma]$ to ψ — is:

- **Increasing in** R_0: higher transmissibility amplifies the effect of punctuation, because more offspring more densely sample the individual infectiousness profile, inheriting more of its temporal structure.
- **Decreasing in** α: more concentrated generation intervals dampen the effect of punctuation, because there is less temporal variation across the generation interval for the shared latent period l_i to exploit.

Together, these results confirm that punctuated infectiousness ($\psi < 1$) has the greatest impact on epidemic timing variability when R_0 is large and the generation interval is dispersed — precisely the regime in which the individual infectiousness profile has the most temporal structure for offspring timing to inherit.

Overdispersion arising from periodic contact patterns

Here, we derive the Negative Binomial overdispersion parameter k [Lloyd-Smith et al., 2005] of the secondary infection distribution under the periodic contact model (Eq. 7), which we restate here in a stationary form with a common random phase $\Phi \sim \text{Uniform}[0, 2\pi)$:

$$c_i(t) = 1 - A\cos(\omega t + \Phi), \qquad A \in [0, 1]. \tag{S23}$$

Randomizing Φ makes $c_i(\cdot)$ stationary and is equivalent to assuming that the infection-time phase t_i mod $2\pi/\omega$ is uniform across individuals, which holds in good approximation during sustained exponential growth.

Lloyd–Smith bridge. The number of secondary infections from person i is $Z_i \sim \text{Poisson}(M_i)$ with $M_i = \int_0^\infty a_i(t_i, \tau)\, d\tau = \nu_i\, \tilde{c}_i$, where

$$\tilde{c}_i := \int_0^\infty b_i(\tau)\, c_i(t_i + \tau)\, d\tau. \tag{S24}$$

Matching the mean and variance of the mixed-Poisson offspring law to those of $\text{NegBin}(R_0, k)$ in the Lloyd-Smith parameterization gives $k = R_0^2/\text{Var}[M_i]$. With $\nu_i \equiv R_0$ held fixed (isolating the contribution of punctuation and contact variability), this reduces to

$$k = \frac{1}{\text{Var}[\tilde{c}_i]}. \tag{S25}$$

The rest of the derivation computes $\text{Var}[\tilde{c}_i]$.

Mean. Conditional on b_i, stationarity gives $E[c_i(t)] = 1$, hence

$$E[\tilde{c}_i \mid b_i] = \int_0^\infty b_i(\tau)\, d\tau = 1, \tag{S26}$$

using that b_i is a probability density. Therefore $E[\tilde{c}_i] = 1$ (and $E[M_i] = R_0$), and by the law of total variance $\text{Var}[\tilde{c}_i] = E\big[\text{Var}[\tilde{c}_i \mid b_i]\big]$.

Autocovariance. Averaging over the uniform random phase Φ,

$$\text{Cov}[c_i(t), c_i(t+s)] = A^2\, E_\Phi[\cos(\omega t + \Phi)\cos(\omega(t+s) + \Phi)] = \frac{A^2}{2}\cos(\omega s), \tag{S27}$$

so $\text{Var}[c_i(t)] = A^2/2$ and the spectrum of the contact process is concentrated at the single frequency ω.

Conditional variance of $\tilde{c}_i$. Substituting Eq. S27 into the double-integral form of the conditional variance,

$$\begin{aligned}
\text{Var}[\tilde{c}_i \mid b_i] &= \int_0^\infty \int_0^\infty b_i(\tau)\, b_i(\tau')\, \frac{A^2}{2}\cos\big(\omega(\tau - \tau')\big)\, d\tau\, d\tau' \\
&= \frac{A^2}{2}\,\text{Re}\left[\left(\int_0^\infty b_i(\tau)\, e^{i\omega\tau}\, d\tau\right)\left(\int_0^\infty b_i(\tau')\, e^{-i\omega\tau'}\, d\tau'\right)\right] \\
&= \frac{A^2}{2}\,\big|\hat{b}_i(\omega)\big|^2,
\end{aligned} \tag{S28}$$

where $\hat{b}_i(\omega) = \int_0^\infty b_i(\tau)\, e^{-i\omega\tau}\, d\tau$ is the Fourier transform (equivalently, the characteristic function at ω) of the biological profile.

Evaluating $|\hat{b}_i(\omega)|^2$. For the time-shifted Gamma model, $b_i(\tau) = f_\epsilon(\tau - l_i)$ with f_ϵ a Gamma$(\psi\alpha, \beta)$ density and random latent period l_i. A time shift multiplies the Fourier transform by a unit-modulus phase, so

$$\hat{b}_i(\omega) = e^{-i\omega l_i}\, \hat{f}_\epsilon(\omega), \qquad \left|\hat{b}_i(\omega)\right|^2 = \left|\hat{f}_\epsilon(\omega)\right|^2, \tag{S29}$$

with the l_i-dependence dropping out entirely. The characteristic function of Gamma$(\psi\alpha, \beta)$ is $\hat{f}_\epsilon(\omega) = (1 - i\omega/\beta)^{-\psi\alpha}$, giving

$$\left|\hat{f}_\epsilon(\omega)\right|^2 = \left(1 + \frac{\omega^2}{\beta^2}\right)^{-\psi\alpha}. \tag{S30}$$

Because Eq. S30 is deterministic, the outer expectation over b_i in the law of total variance is trivial:

$$\mathrm{Var}[\tilde{c}_i] = \frac{A^2}{2}\left(1 + \frac{\omega^2}{\beta^2}\right)^{-\psi\alpha}. \tag{S31}$$

Closed-form expression for k. Combining Eqs. S25 and S31,

$$\boxed{k = \frac{2}{A^2}\left(1 + \frac{\omega^2}{\beta^2}\right)^{\psi\alpha}.} \tag{S32}$$

The expression has two factors with transparent interpretations. The prefactor $2/A^2$ is the reciprocal variance of a uniformly-phased sinusoid with amplitude A, and it sets the overall scale: larger contact-oscillation amplitude $\Rightarrow$ smaller k $\Rightarrow$ more overdispersion. The bracketed factor $(1 + \omega^2/\beta^2)^{\psi\alpha}$ is the reciprocal of the squared characteristic function of b_i at the contact frequency ω, and it quantifies how much the biological profile's temporal window fails to average out the contact oscillation. This factor increases with ψ for any $\omega > 0$, so narrower profiles always produce more overdispersion in the presence of any periodic contact variation.

Limiting cases.

- **Spike profile, $\psi \to 0$:** the bracketed factor becomes 1, leaving $k = 2/A^2$. A spike profile samples c_i at a single instant, so the conditional variance of $\tilde{c}_i$ equals $\mathrm{Var}[c_i] = A^2/2$ regardless of ω or β.
- **Smooth profile, $\psi \to 1$:** $k = (2/A^2)(1 + \omega^2/\beta^2)^\alpha$, strictly larger than the spike value whenever $\omega > 0$.
- **Slow oscillation, $\omega/\beta \to 0$:** the contact process looks locally constant over the infection window, so $\tilde{c}_i \approx c_i(t_i)$ for any ψ, and $k \to 2/A^2$ independent of ψ.
- **Fast oscillation, $\omega/\beta \to \infty$:** the contact process averages out over any infection window with $\psi > 0$, driving $\mathrm{Var}[\tilde{c}_i] \to 0$ and $k \to \infty$ (Poisson).
- **No contact variation, $A \to 0$:** $k \to \infty$ trivially.

These limits are consistent with the intuition that overdispersion arises from incomplete averaging of contact variation by the biological profile window, and that the degree of averaging is controlled by both $\psi\alpha$ (profile width) and ω/β (ratio of profile timescale to contact timescale).

.0.1 *Renewal equation model*

Early work by Hudson, and later by Kermack and McKendrick [Kermack and McKendrick, 1927], modeled the transmission of a permanently immunizing infection using an integral equation. In modern notation [Breda et al., 2012], the force of infection is

$$F(t) = \int_0^\infty F(t-\tau)S(t-\tau)A(\tau)d\tau \tag{S33}$$

where $F(t)$ is the force of infection at time t, $S(t)$ is the density of susceptible individuals in the population at time t, and $A(\tau)$ is the infectiousness profile describing the expected contribution to the force of infection from an individual who was infected τ time units ago. The incidence of disease is the product of the force of infection and the susceptible density (*i.e.*, $-\frac{dS}{dt} = F(t)S(t)$); thus, the infectiousness profile $A(\tau)$ is the fundamental object that determines how an epidemic unfolds.

Supplementary Figures

Supplementary Tables